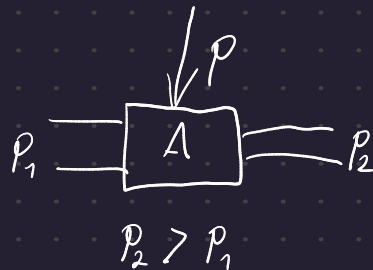


1. Amplifiers

power at input < power at output



✗ transformer is NOT an amplifier

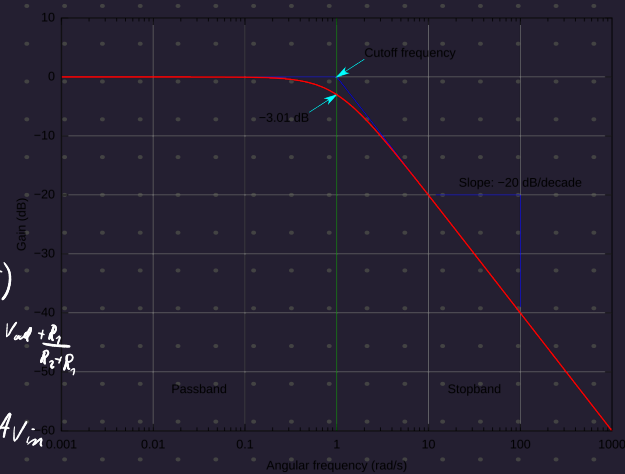
- BJT
- MOS
- Op-Amp



$$V_{out} = A(V^+ - V^-)$$

$$V^- = V_{in} \frac{R_f}{R_i + R_f}$$

$$V_{out} \left(1 + A \frac{R_f}{R_i + R_f} \right) = A V_{in}$$



Frequency response of Low-pass filter

Frequency response concerns both
Amps and filters

2. Filters